

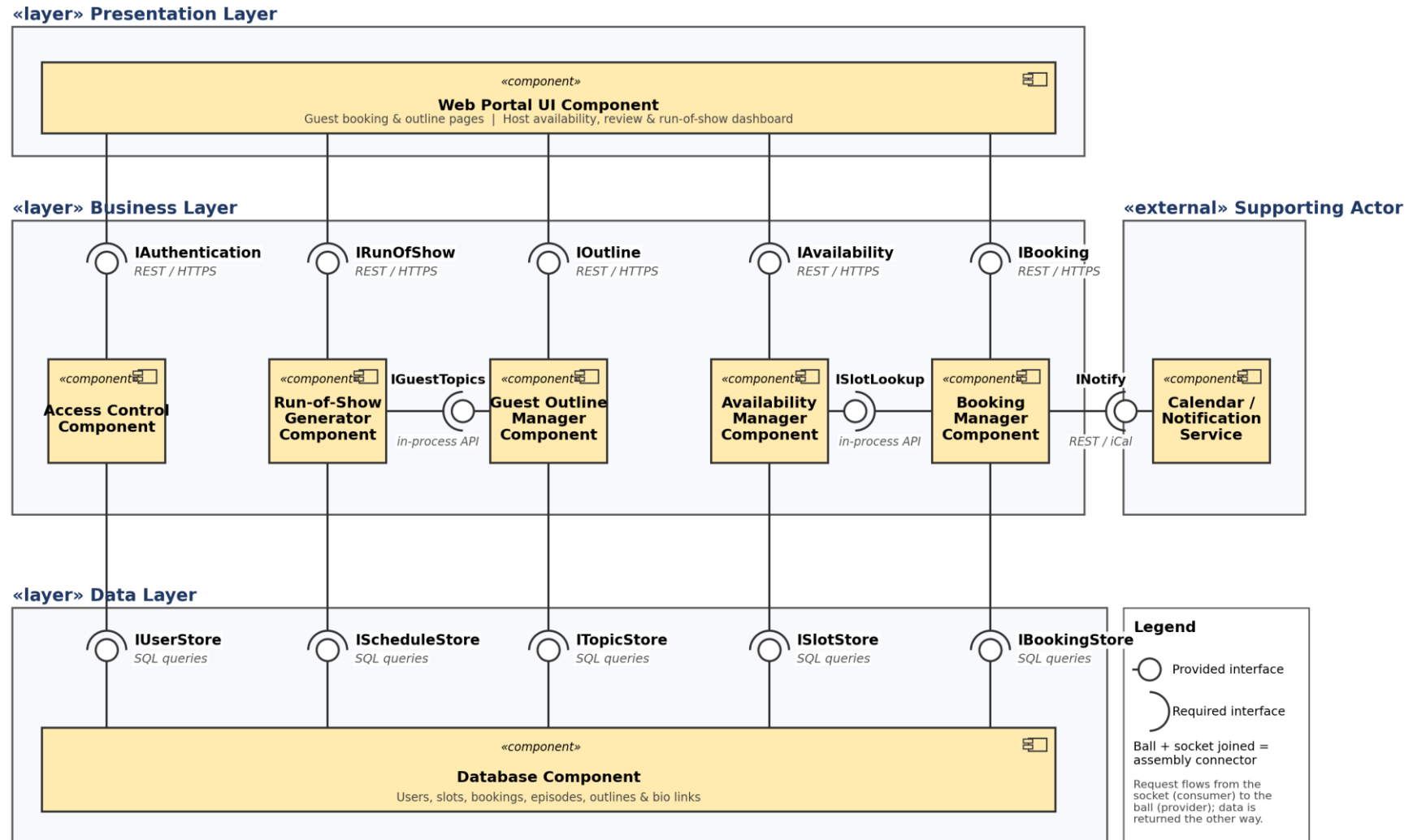
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Lab 3 – Deliverable 1: Component Diagram

Problem Statement #60 – Podcast Guest Scheduling & Outline Builder

Podcast Guest Scheduling & Outline Builder - Layered Architecture (UML Component Diagram)



Lab 3 – Deliverable 2: Written Justification

Architecture Selection: *We chose a Layered Architecture for the Podcast Guest Scheduling & Outline Builder System.*

Architectural Choice

Layered (three-tier) architecture with a Presentation Layer (Web Portal UI), a Business Layer (Access Control, Availability Manager, Booking Manager, Guest Outline Manager and Run-of-Show Generator components) and a Data Layer (Database Component). The Calendar/Notification Service is the only external component. Each layer talks to the one below it only through provided/required interfaces (e.g., IBooking, ISlotLookup, IBookingStore).

Two Reasons for This Choice

- **The requirements split cleanly into layers.** Guest and host screens for booking, availability and outline entry (FR-001, FR-002, FR-003) belong to the presentation layer; slot rules, booking, outline compilation and run-of-show generation (FR-001 to FR-005) are business logic; slots, bookings, outlines and bio links are persistent data. Because every FR maps to one layer and one component, the work is easy to divide, build and test independently.
- **The system is small, single-organisation and needs strong consistency.** Only two actors (Podcast Guest, Show Host) use it, and the workflow is tightly linked: book a slot, submit an outline, generate the run-of-show. A single central database makes it simple to guarantee that a blocked or already-booked slot can never be booked (the FR-001/FR-002 fail criteria). Microservices would add network latency, operational complexity and distributed-data consistency problems that this scale does not justify, and plain client-server offers no internal structure to keep logic separate from storage.

Security Advantage

The UI never touches the database directly. Every request must pass through the Access Control Component and the business-layer interfaces, so authentication and authorisation are enforced in one place. This directly supports NFR-002: a guest can reach only their own booking and outline, and a host only the episodes they manage. Input such as topics and biography links is validated in the business layer before it is stored, and the Data Layer can be placed on a private network that only the business layer can reach.

Performance Benefit

The business-layer components (Run-of-Show Generator, Guest Outline Manager, Booking Manager) are deployed together and call each other in-process, so there are no network hops between them. The Run-of-Show Generator can fetch the episode schedule and guest topics through short indexed queries and render the PDF production sheet inside the 1-second target of NFR-001. Slot and booking lookups can also be cached in the business layer, keeping the booking page responsive when several guests view availability at once.

Trade-off Acknowledged

Layering adds a small call overhead and couples the layers through their interfaces. At this system's size the overhead is negligible, and the explicit interfaces make each component replaceable (for example, swapping the Calendar/Notification Service) without changing the rest of the system.

Appendix: Supporting Analysis (Lab 3 Steps 1–3)

This appendix records the working behind the diagram and can be removed from the submitted justification if only the one-page version is required.

Step 1 – Scenario Review

- **Key functional needs:** slot booking and outline submission (FR-001), host slot management (FR-002), host review of submissions (FR-003), run-of-show PDF generation (FR-004), booking confirmation with calendar invite (FR-005).
- **Key non-functional needs:** production sheet generated in under 1 second (NFR-001); only authorised users may access episode data (NFR-002).
- **Main challenges:** preventing double-booking or booking blocked slots; protecting guest information; PDF latency; reliability of the external calendar service; a simple outline form for guests.

Step 2 – Architectural Style Analysis

Style	Pros for this scenario	Cons for this scenario	Verdict
Layered	Clear separation of UI, logic and data; simple to build and test; easy consistency with one database; matches the FR structure.	Some call overhead; layers are coupled; harder to scale one component alone.	Selected
Microservices	Independent scaling and deployment; fault isolation.	Operational complexity, network latency and data-consistency problems; excessive for two actors and one organisation.	Rejected
Client-Server	Centralised control; simple deployment; easy data consistency.	Single point of failure; no internal structure, so UI, rules and data logic tend to blend together.	Rejected

Step 3 – Components Identified

Component	Layer	Responsibility	Requirements
Web Portal UI Component	Presentation	Guest booking and outline pages; host availability, review and run-of-show dashboard.	FR-001 – FR-004
Access Control Component	Business	Authenticates users and enforces guest/host permissions on episodes and submissions.	NFR-002
Availability Manager Component	Business	Creates, modifies and blocks interview slots; answers slot-availability lookups.	FR-002
Booking Manager Component	Business	Verifies slot is free, saves the booking, marks the slot unavailable, triggers confirmation (UC-02, UC-06).	FR-001, FR-005
Guest Outline Manager Component	Business	Stores discussion topics and biography links; lets the host review them.	FR-001, FR-003
Run-of-Show Generator Component	Business	Builds the timestamped PDF production sheet from the schedule and guest topics.	FR-004, NFR-001
Database Component	Data	Persists users, slots, bookings, episodes, outlines and bio links.	All
Calendar / Notification Service	External	Sends booking confirmation and calendar invitation to the guest (supporting actor).	FR-005

Interface Summary

Interface	Provided by	Required by	Data flow / technology
IAuthentication	Access Control	Web Portal UI	Login / role check requests; returns session and permissions (REST/HTTPS)
IAvailability	Availability Manager	Web Portal UI	Host slot create/modify/block; guest list of open slots (REST/HTTPS)
IBooking	Booking Manager	Web Portal UI	Guest slot selection and confirmation; returns booking result (REST/HTTPS)

Interface	Provided by	Required by	Data flow / technology
IOutline	Guest Outline Manager	Web Portal UI	Submit/view topics and bio links (REST/HTTPS)
IRunOfShow	Run-of-Show Generator	Web Portal UI	Host requests production sheet; returns PDF (REST/HTTPS)
ISlotLookup	Availability Manager	Booking Manager	Check and mark slot availability (in-process API)
IGuestTopics	Guest Outline Manager	Run-of-Show Generator	Fetch topics for an episode (in-process API)
INotify	Calendar/Notification Service	Booking Manager	Send confirmation + calendar invite (REST / iCal)
IUserStore	Database	Access Control	User and role queries (SQL)
ISlotStore	Database	Availability Manager	Slot read/write (SQL)
IBookingStore	Database	Booking Manager	Booking and episode read/write (SQL)
ITopicStore	Database	Guest Outline Manager	Outline and bio link read/write (SQL)
IScheduleStore	Database	Run-of-Show Generator	Episode schedule read (SQL)